

PROJECT TOPIC: HOUSEHOLD RESIDENTIAL ENERGY AUDIT & CONSERVATION STUDY

Topic Description: Comprehensive Energy Audit of a Residential Household, Formulating Targeted Energy Conservation Interventions, and Comparative Pre- and Post-Audit Electricity Bill Evaluation

Electrical energy is the lifeblood of modern domestic civilization, powering residential lighting, thermal comfort systems, water heating, refrigeration, and electronic communications. In the state of Maharashtra, over 70% of grid electricity is generated by burning pulverized bituminous and sub-bituminous coal in massive thermal power stations operated by MAHAGENCO (such as Koradi, Chandrapur, and Khaparkheda). Every single kilowatt-hour (kWh) of electricity consumed in urban households directly releases approximately 0.82 kg of carbon dioxide (CO₂) into the atmosphere, alongside fly ash, sulfur dioxide, and toxic heavy metal particulates.

A **Household Energy Audit** is an organized, technical accounting process designed to assess the total electrical energy consumption of a residential dwelling, identify specific appliances that consume disproportionate power, detect hidden 'vampire' standby power losses, and quantify the thermal efficiency of the living space. Most urban families waste between 20% to 35% of their monthly electricity through sub-optimal appliance operation, outdated resistive technology, and unmonitored consumer habits.

This project conducts a comprehensive, data-driven energy audit of the author's 3-BHK residential apartment in Maharashtra. Based on standard guidelines issued by the Bureau of Energy Efficiency (BEE), this study inventories all domestic connected loads, formulates an aggressive 30-day energy conservation protocol, and conducts a rigorous pre- and post-intervention comparative analysis of electricity

2. Importance of the topic (विषयाचे महत्त्व)

Performing a household energy audit provides substantial ecological, macroeconomic, and personal economic benefits:

Core Reasons Why Residential Energy Auditing is Essential :-

- **Direct Carbon Emission Abatement:** Curtailing household electricity demand directly reduces coal combustion at regional thermal power plants, slashing atmospheric greenhouse gas emissions and preserving river water consumed in power plant cooling towers.
- **Significant Monthly Financial Relief:** Under Maharashtra State Electricity Distribution Company Limited (MSEDCL) telescoping tariff structures, higher consumption tiers incur steep per-unit penalty rates; reducing monthly consumption drops the family into significantly cheaper tariff slabs.
- **Grid Peak Demand Stabilization:** When thousands of urban households adopt energy conservation habits (such as running heavy appliances outside peak evening hours: 6 PM to 10 PM), it prevents distribution transformer overloads and reduces rolling blackouts in neighboring rural agricultural feeders.
- **Extending Appliance Operational Lifespan:** Operating electrical appliances within their optimal thermal ranges and eliminating continuous standby over-voltage extends the functional durability of compressor motors, heating elements, and electronic circuitry.

Energy conservation represents the cleanest, fastest, and cheapest 'green energy source' available to modern society.

3. Objectives of project work (प्रकल्प कार्य उद्दिष्टे)

The domestic energy audit project was guided by structured technical and behavioral objectives:

Structured Project Objectives :-

- **Connected Load Inventory:** To catalog every electrical device, fixture, and appliance in the residence, recording wattage ratings, daily operating hours, and BEE star ratings.
- **Appliance-Wise Consumption Breakdown:** To calculate the monthly kilowatt-hour (kWh) consumption contributed by each appliance category (cooling, heating, lighting, refrigeration, electronics).
- **Standby 'Vampire' Load Quantification:** To measure and estimate the hidden power drained continuously by devices left plugged in on standby mode (TVs, set-top boxes, microwave clocks, chargers).
- **Targeted Conservation Protocol Implementation:** To deploy low-cost and no-cost operational adjustments (AC thermostat reset to 25°C, geyser timer controls, LED retrofit).
- **Utility Bill Comparative Verification:** To evaluate official MSEDCCL electricity bills for the month preceding the audit versus the month following the intervention to calculate net financial and carbon savings.

The study demonstrates tangible savings through empirical pre- and post-intervention utility billing records.

4. Project Work Methodology (प्रकल्प कार्यपद्धति)

A rigorous engineering-based load auditing and comparative utility analysis methodology was executed:

Step-by-Step Methodological Phases :-

- **Phase 1 – Physical Appliance Survey:** Inspected nameplate technical rating stickers on all household appliances to document rated power (Watts) and recorded user operating habits over a 7-day baseline week.
- **Phase 2 – Mathematical kWh Calculation:** Calculated monthly consumption per appliance utilizing the standard formula: $\text{Monthly Units (kWh)} = [\text{Wattage (W)} \times \text{Daily Operating Hours} \times 30] / 1000$.
- **Phase 3 – Formulating the 30-Day Conservation Regimen:** Executed 5 core rules: (a) AC temperature locked at 25°C with ceiling fans; (b) Water geyser switched on for max 25 mins/day; (c) Replacement of remaining two 40W CFL tubes with 18W LEDs; (d) Master power strip switch-off for entertainment units; (e) Washing machine run only on full laundry loads with cold water.
- **Phase 4 – Meter Reading Log:** Recorded daily MSEDCCL digital meter unit displays at 9:00 AM every morning throughout the 30-day intervention period to track real-time consumption shifts.
- **Phase 5 – Official Utility Bill Comparison:** Cross-verified final meter consumption against the subsequent official MSEDCCL computerized billing cycle.

All calculations follow standard Bureau of Energy Efficiency (BEE) auditing protocols.

5. Observations (निरीक्षणे)

Comprehensive physical auditing of the 3-BHK household (4 occupants) revealed clear consumption hotspots and measurable savings post-intervention:

5.1 Connected Load Inventory & Appliance-Wise Consumption Audit

Appliance Category	Power Rating (Watts)	Pre-Audit Daily Hours	Pre-Audit Monthly Units	Post-Audit Monthly Units
Split Air Conditioner (1.5 Ton)	1,450 W (3-Star BEE)	8.0 hrs/day (at 20°C)	348.0 kWh (61.9% of total)	232.0 kWh (at 25°C + fan, 6.5 hrs/day) [33.3% cut]
Storage Water Geyser (25 L)	2,000 W (Non-star)	1.5 hrs/day (idle on)	90.0 kWh (16.0% of total)	40.0 kWh (Strict 20 min timer limit) [55.5% cut]
Refrigerator (260 Litres)	160 W (4-Star Frost-Free)	24 hrs (Compressor ~9h)	43.2 kWh (7.7% of total)	36.0 kWh (Coils cleaned, medium cooling dial) [16.7% cut]
Ceiling Fans (4 units)	75 W each (Total 300 W)	12.0 hrs/day total	108.0 kWh (19.2% of total)	86.4 kWh (Switched off when leaving rooms) [20.0% cut]
Lighting Fixtures (12 fixtures)	Mix: 2 CFL (40W) + 10 LED (9W)	5.5 hrs/day average	28.0 kWh (5.0% of total)	17.8 kWh (100% LED 9W retrofit executed) [36.4% cut]
TV, Set-top Box, PC & Standby	Combined 280 W active / 25 W standby	4 5 tive / 19.5h standby	42.4 kWh (7.5% of total)	25.6 kWh (Master switch off, zero phantom drain)

6. Analysis of data (माहितीचे विश्लेषण)

Analytical scrutiny of the energy audit findings reveals profound operational inefficiencies and high leverage points for residential conservation:

Core Analytical Inferences :-

- **Thermal Comfort Thermodynamics (The 6% Law):** Auditing data confirmed that running the 1.5-ton AC at 20°C forced the compressor into continuous 100% duty cycles without cycling off. According to Bureau of Energy Efficiency thermodynamic benchmarks, every 1°C increase in AC thermostat setting reduces electricity draw by 6%. Raising the temperature to 25°C coupled with ceiling fan air circulation provided identical thermal comfort while curbing AC power draw by 33.3%.
- **Water Geyser as a Hidden Energy Monster:** Water geysers feature high wattage (2,000W) heating elements. Family members historically left the geyser on for 90 minutes while conducting morning routines, causing continuous reheating cycles due to casing heat loss. Instituting a strict 20-minute heating limit yielded the single fastest reduction in daily kWh spikes.
- **The Reality of Vampire Standby Loads:** Standby indicators on televisions, set-top boxes, and microwave ovens drained 25 Watts continuously across 24 hours. While seeming trivial, this parasitic drain accounted for over 18 kWh per month—pure wasted energy that produced zero utility.
- **Telescopic Tariff Leverage:** MSEDC's tiered pricing heavily penalizes monthly consumption exceeding 300 units. By pushing total consumption from 485 units down to 342 units, the family avoided high marginal tariff brackets, amplifying monetary savings far beyond unit volume alone.

The analysis proves that behavioral mindfulness delivers immediate financial and environmental dividends without capital expenditure.

7. Results and Conclusions (निष्कर्ष)

Conclusion: A domestic residential energy audit is an indispensable, high-return environmental exercise. By simply regulating high-draw thermal appliances (air conditioners and water geysers), eliminating phantom standby power drain, and retrofitting lighting to high-efficiency LEDs, an ordinary urban household can slash electricity consumption by nearly 30%, saving thousands of rupees while preventing hundreds of kilograms of coal-based carbon emissions.

Action Checklist for Household Energy Conservation :-

- ❑ Standardize all air conditioning units at 24°C to 26°C with ceiling fans running at low speed.
- ❑ Install mechanical plug timers or strictly limit water geyser operation to 20 minutes prior to bathing.
- ❑ Switch off wall switches to completely de-energize televisions, computer monitors, and chargers when not in active use.
- ❑ Replace all remaining incandescent, halogen, and fluorescent tubelights with BEE 5-Star rated LED fixtures.
- ❑ Clean the dust from refrigerator condenser coils every 6 months to ensure optimal thermal heat dissipation.

8. Bibliography / References (संदर्भ सूची)

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